

Supplemental Materials

Table S1. Summary statistics of the training datasets and standardized gene annotations.

Metric	Bacteriophage	Prokaryote
Total genomes	16,479	33,469
Minimum genome length	1.79 kb	207.81 kb
Median genome length	44.83 kb	4.3 Mb
Maximum genome length	734.31 kb	14.2 Mb
Total genera	1,397	4,900
Total proteins	1,473,732	142,978,349
Median proteins per genome	66	3,947
Hypothetical / dark proteins (%)	60.2	12.7
PHROG annotation (%)	73.7	-
Pfam annotation (%)	-	90.5
COG annotation (%)	-	92.6
eggNOG OG annotation (%)	-	90.3
Minimum proteins per vocabulary term	50	50
Median proteins per vocabulary term	270	1,029
Maximum proteins per vocabulary term	19,930	2,069,836
Standardized vocabulary size	582	11,999
OTHER token (%)	0.82	0.106

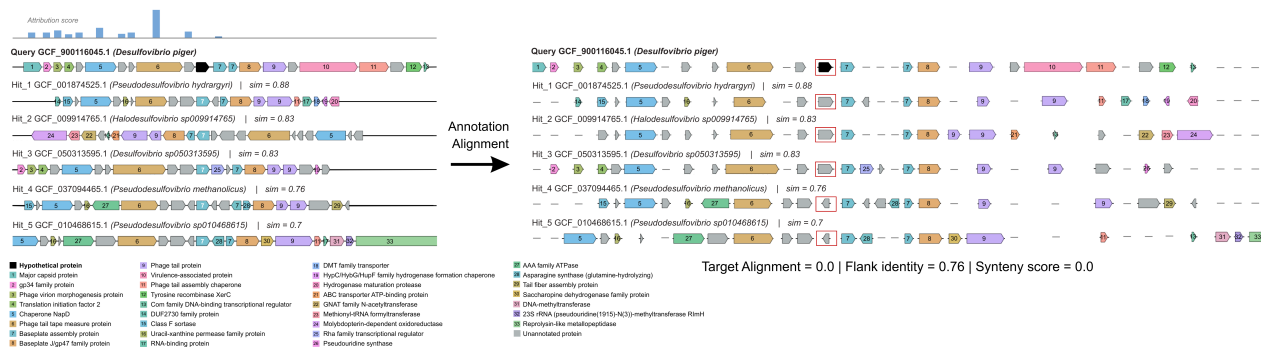


Figure S1. Example of post-hoc annotation alignment of a “context-leakage” prediction. The left panel shows neighbor attribution along with annotations of top 5 retrieved context neighborhoods pre annotation alignment, with the target gene predicted to be “Baseplate assembly protein”. The right panel shows post-alignment results, with the target gene aligned with unannotated genes in the retrieved examples, resulting in low target alignment and synteny score.

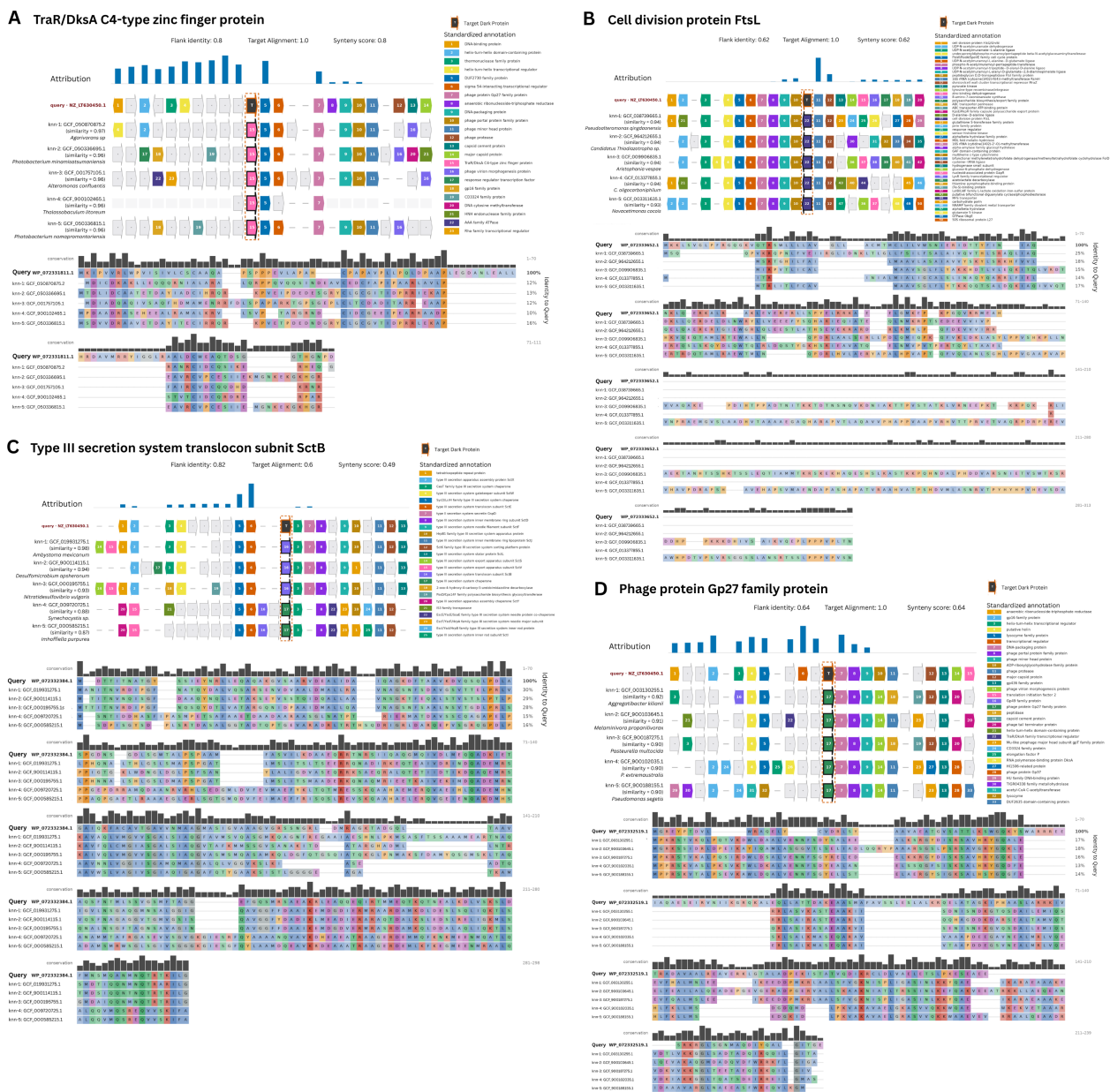


Figure S2. Multiple sequence alignment and Interpretation panels of Synnotate-annotated genes. (A) TraR/DksA C4-type zinc finger protein. (B) Cell division protein FtsL. (C) Type III secretion system translocon subunit SctB. (D) Phage protein Gp27 family protein. Protein multiple sequence alignments were generated using MAFFT³³ and visualized with pyMSAviz³⁴.

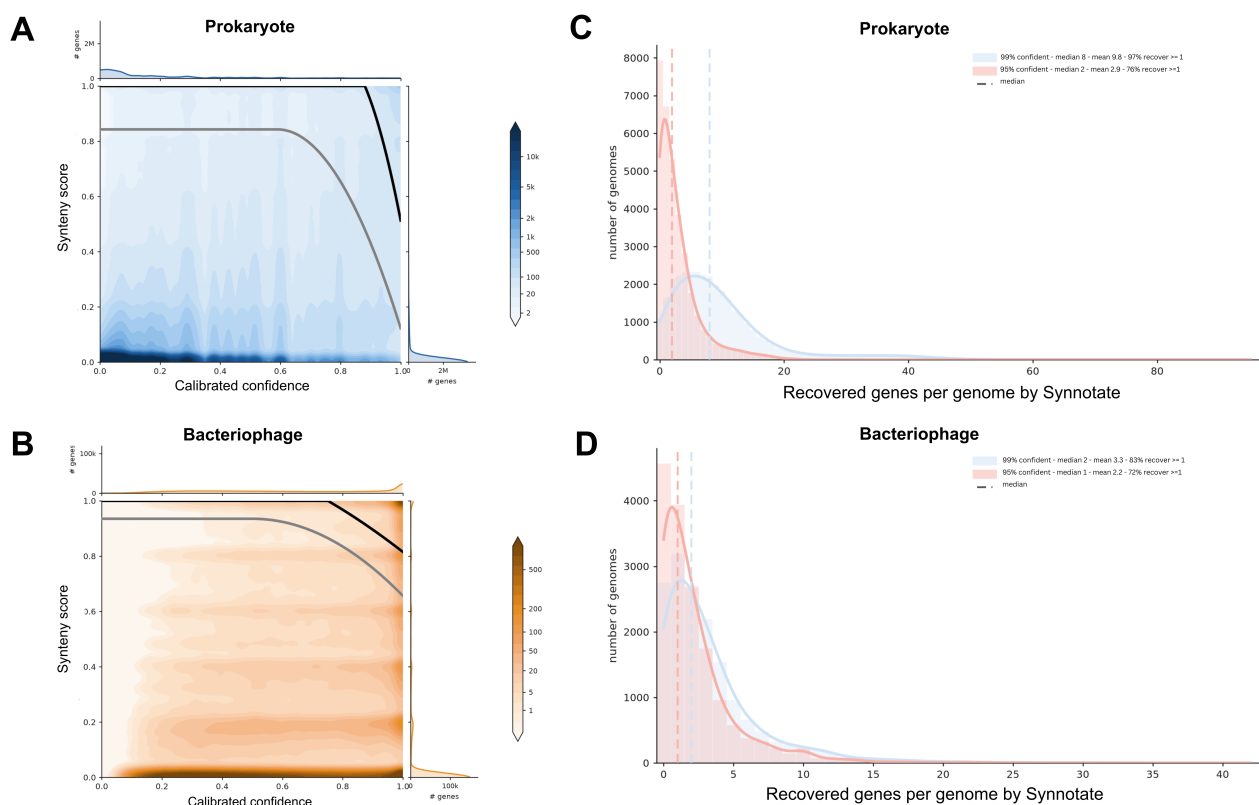


Figure S3. Annotation of the dark proteome of prokaryotic and bacteriophage datasets. (A) Synnotate calibrated confidence and Synteny score of dark genes in Prokaryote dataset. (B) Synnotate calibrated confidence and Synteny score of dark genes in Bacteriophage dataset. (C) Number of genes confidently annotated by Synnotate per genome in Prokaryote dataset. (D) Number of genes confidently annotated by Synnotate per genome in Bacteriophage dataset.

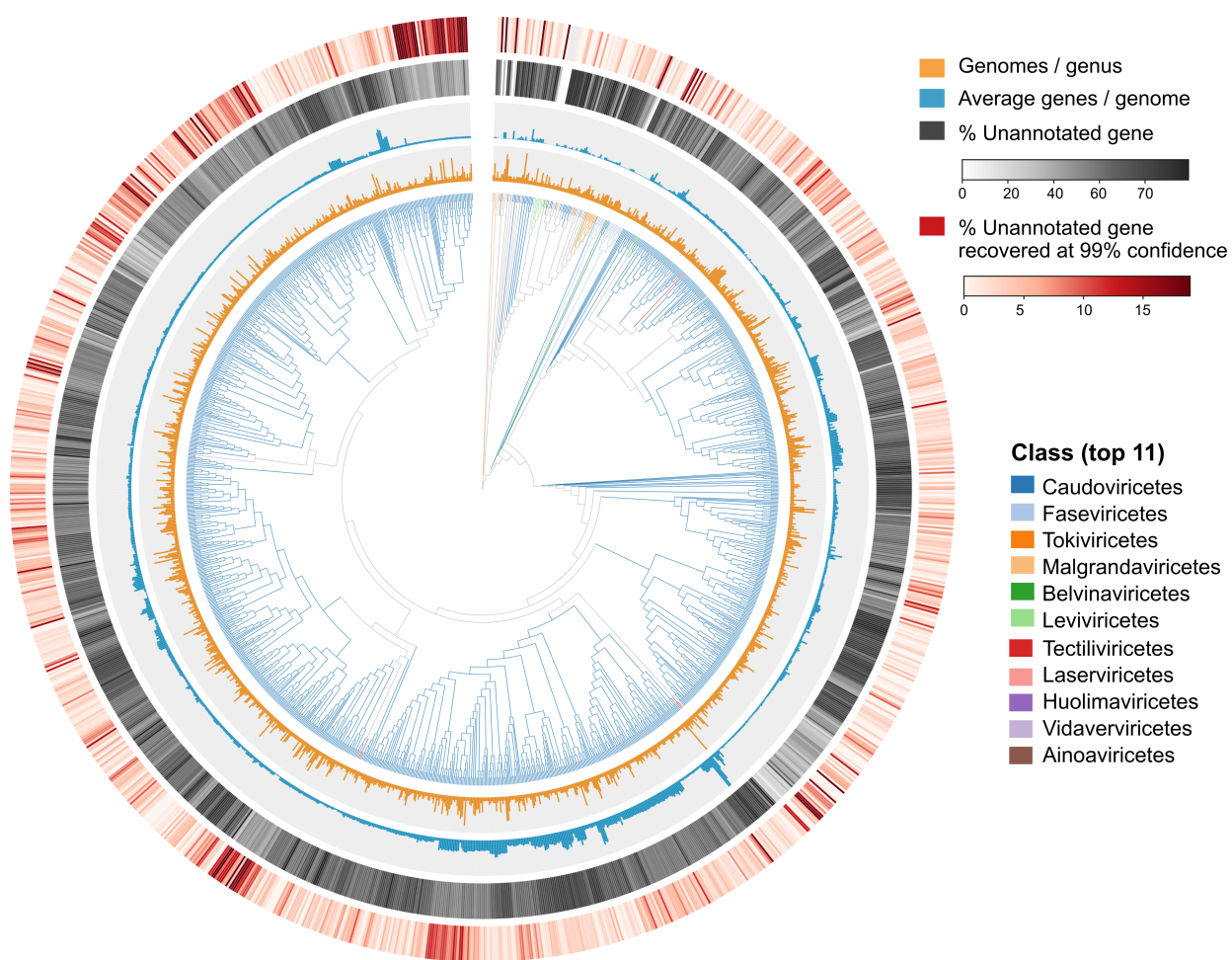


Figure S4. Distribution of dark proteome and Synnotate-annotated genes across viral genera.